Module 1
Learn With PsycLearn Essentials

Module 2
Opening to Depressive and Bipolar Disorders

Module 3
Depressive and Bipolar Disorders

Module 4
Biological Contributing Factors

Lesson 5: Genetic Contributing Factors

- Developmental Psychopathology — Reading • 3 min
- Genetic Contributing Factors — Reading • 3 min
- The Link Between Genes and Psychopathology — Video • 4 min
- Practical Scenario — Real-world application • 5 min
- Genetic Contributing Factors for Depression — Reading • 10 min
- Epigenetic Contributors to Depression — Video • 3 min
- Genetic Contributing Factors for Bipolar Disorder — Reading • 3 min
- Check Your Understanding: Genetic Contributing Factors — Practice Assignment • 4 min

Lesson 6: Neurobiological Contributing Factors

- Depressive Disorders: A Case Study — Reading • 5 min
- Communication in the Synapse — Video • 4 min
- Neurochemical Contributors to Depression — Reading • 10 min
- Practical Scenario — Real-world application • 5 min
- Hormonal Contributors to Depression — Reading • 5 min
- Brain Regions Linked to Depression — Video • 4 min
- Neurobiological Contributors to Depressive and Bipolar Disorders — Reading • 15 min
- Check Your Understanding: Biological Contributing Factors — Practice Assignment • Grade: 83.33%

Key Takeaways

- Key Takeaways: Biological Contributing Factors — Reading • 10 min

Module 4 Quiz

- Module Quiz: Biological Contributing Factors — Graded Assignment • 9 min

Module 5
Psychological Contributing Factors

Module 6
Treatments for Depressive and Bipolar Disorders

Module 7
Suicide

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Conclusion

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# Key Takeaways: Biological Contributing Factors

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## Learning Objectives

This module focused on helping you to be able to:

- Compare and contrast common mood disorders.
  - Explain the genetic contributing factors for mood disorders.
  - Discuss the neurobiological contributing factors for mood disorders.

**Learning objective: Explain the genetic contributing factors for mood disorders.**

1. There is an interplay among biological or psychological factors. Contemporary researchers and clinicians use a **developmental psychopathology** approach to examine the origins and progression of psychological disorders by focusing on the typical processes of human growth and maturation.
2. At every stage of development, we experience biological, social, cognitive, emotional, and psychological challenges. All of these factors should be considered in explaining behavior; no single factor in isolation is responsible for causing psychological disorders.
3. The concept of **equifinality** describes how different experiences over the lifespan can contribute to the development of the same outcome, in this case a specific psychological disorder (Almy & Cicchetti, 2018).
4. The concept of **multifinality** describes how similar experiences during development end up with different outcomes. The presence of a particular factor during development does not necessarily lead to the same outcome for every individual; instead, the same factors may play out differently for different people.
5. The developmental psychopathology approach recognizes that multiple factors may contribute to how psychological disorders develop. These include biological and psychological factors. **Biological factors** include genetics, neural activity, hormones and neurochemical levels, and environmental factors such as social and cultural influences. **Psychological factors** include cognitive distortions, level of stress, and ruminating behaviors.
6. **Heritability** is a mathematical calculation of the degree of variability in our genes. It is used to estimate the variability in the expression of a physical trait, such as height or eye color, as well as psychological disorders, such as depression. Heritability is a statistical estimate of the contribution of both inheritance and environment to a given trait in a population, but not in a particular individual.
7. Family and twin studies of depression have led to an estimate of a heritability rate between 30% and 50% among identical twins. Identical twins share 100% of their genes, whereas fraternal twins share only 50% of their genes. When one identical twin has a depressive disorder there is an increased risk that the other twin will also develop depressive symptoms (McIntosh et al., 2019).
8. There are genetic predispositions to bipolar disorder. Bipolar disorders and depressive disorders have strong genetic links. Of the major psychological disorders, bipolar disorders are among the ones with the highest heritabilities.
9. The **concordance rate** is the percentage of twins or other blood relatives who exhibit a particular trait or disorder. If siblings exhibit the same trait, they are said to be concordant, and if they do not exhibit the same trait, they are said to be discordant. There is a genetic component to depression. The concordance rate is higher among identical twins than fraternal twins.
10. The heritability of bipolar disorder is 70%. Concordance rates in twin studies are high, if one identical twin is diagnosed with bipolar disorder there is a 40-70% chance the other twin will be diagnosed as well. The lifetime risk of developing bipolar disorder in first-degree relatives of bipolar patients is as high as 10%, far higher than the 1.5% risk for the general population.
11. Relatives of bipolar patients are more likely to experience higher rates of other psychological disorders. In one study, the rate of psychological disorders in first-degree relatives of individuals with bipolar disorder was 15% for individuals with bipolar I disorder and 26% for individuals with bipolar II disorder. Relatives of individuals with bipolar II disorder were also more likely to be diagnosed with severe depression.
12. **Epigenetics** is the field of study that investigates the ways in which the environment can lead to the activation and deactivation of various parts of our genome. Environmental influences, such as stress, can influence the expression of genes, and can increase or decrease the likelihood that a person will develop a physical or psychological disorder. Maltreatment during childhood, for example, is an environmental factor that has been found to activate parts of the genome related to depression and suicidal ideation. Epigenetics is one mechanism to explain how environmental influences can affect only one member of a twin pair.

**Learning objective: Discuss the neurobiological contributing factors for mood disorders.**

1. The primary neurotransmitters related to depression are serotonin, noradrenaline, glutamate, and gamma-aminobutyric-acid (GABA). Serotonin, noradrenaline, and glutamate are excitatory neurotransmitters involved in mood. When the levels of these neurotransmitters are low, as they are in individuals with depression, mood is low. The major psychotropic medications for depression are designed to increase levels of these neurotransmitters in the brain, thereby increasing mood. GABA, by contrast, is an inhibitory neurotransmitter, whose effect is to calm brain activity.
2. Research suggests that the sex hormones—estrogen, progesterone, and the androgens (which include testosterone)—play a role in increasing the risk for depression. This seems to be particularly true among women, which may account for the consistent and striking gender difference in depression, and explain why there is a correlation between hormonal changes and depression across the lifespan.
3. The majority of hormones that impact psychological health, including depression, are tied to stress. When a person is under stress, the body mobilizes to initiate a stress response, triggering a multi-step chain of events that ultimately leads to the release of **adrenaline**, also called epinephrine, a hormone that is secreted in large amounts in response to fear, anxiety, or stress, and the stress hormone cortisol.
4. **Cortisol** is released and improves the body's ability to manage stress and to repair itself. However, exposure to prolonged or chronic stress can lead to excessive levels of cortisol, which can have harmful effects on the body. The experience of chronic stress and chronically elevated levels of cortisol are associated with increases in depression.
5. Localizing a single brain region as a contributor to depression is not possible because many brain regions are implicated in depression. Recent research supports the hypothesis that depression is more likely due to dysfunction of the brain circuits that link multiple brain regions rather than any one brain region.
6. Bipolar disorder results from the summation of many genetic variances that predispose people to develop the disorder.
7. Researchers have observed atypical activity in numerous brain regions including the prefrontal cortex, the subgenual anterior cingulate cortex, the amygdala, the ventral striatum, and the hippocampus of patients with symptoms of depression.
8. Bipolar disorder doesn't arise out of a single brain area but rather it is associated with dysfunctions in brain circuits between many interconnected areas. Regions implicated in bipolar disorder include the anterior cingulate cortex, orbitofrontal cortex, ventral striatum, hippocampus, amygdala, and insula.
9. Brain tissue loss, such as in the anterior cingulate cortex, the orbitofrontal cortex, the ventral striatum, the hippocampus, and the insula, or brain tissue gain, such as in the amygdala, are associated with bipolar disorders. Other research has found decreased activity, such as in the anterior cingulate cortex and ventral striatum, which are involved with regulating emotions and with the reward circuitry in the brain that regulates feelings of pleasure. Increased activity in brain areas such as the orbitofrontal cortex, amygdala, and insula, have also been linked to bipolar disorder.
10. Most of this work has focused on the functioning of brain structures closely linked to emotional processing. This larger "limbic network" shows both functional and structural differences in people with bipolar disorder. The neural activity in these areas is different in people with bipolar disorder and those without, as are the physical attributes of the areas, such as their size and number of neural connections.

Now that you've finished reviewing these key takeaways, you can go back and look over the material from this module or move on to take the Module Quiz. Remember to use the information you just learned when you study for the quiz. Good luck!

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